Counterexample-Guided Short-Horizon Shielding of Generative Robot Policies

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Abstract. Generative robot policies commit to several future motions at once. An unfamiliar obstacle can therefore cause a collision before the next policy query. We present ShortStop, a runtime shield that checks and repairs these action chunks without retraining the policy. Bands for model error around predicted arm poses form a reachtube. A lower bound on Signal Temporal Logic (STL) robustness certifies chunks whose tube–obstacle clearance meets a margin. If all candidates fail, the point of least clearance guides repair, with joint holding as fallback. Safety depends on tube coverage, for which we derive a statistical bound. Inference delay requires a certified horizon $H \geq r + d$, where r steps execute per decision and computation takes d steps. On CALVIN (Composing Actions from Language and Vision), a frozen Multimodal Diffusion Transformer with our shield reduces collisions from 39.7% to 1.1% across 1000 pre-registered obstacle scenes. Collision-free task completion rises from 58.4% to 80.0%, comparable to filters sharing our model and margins. The arm leaves its tube in 0.10% of 41,899 audited steps. With three steps of inference delay, the extended horizon yields 0.0% violations, compared with 31.5% without delay awareness.

Keywords: Safe robot learning · Diffusion policy · Action chunking · Reachability · Signal temporal logic · Runtime shielding.

1 Introduction

A robot reaching for a drawer handle may encounter a ball that was absent from its demonstrations (Fig. 1). If the policy cannot see the ball, it proposes the familiar approach. Holding the arm avoids a collision but leaves the drawer closed. To finish the task, the robot must adjust that approach while preserving its progress toward the handle.

Diffusion models [11, 19] and flow matching [22] trained on demonstrations represent several possible motions for one observation. Both emit *action chunks*, sequences of future actions whose prefix runs before the next policy query. Chunking improves temporal consistency and throughput [11], but the executed prefix is *open-loop*: the policy cannot revise it in response to an obstacle. Verifying every possible output would require reasoning through repeated generative

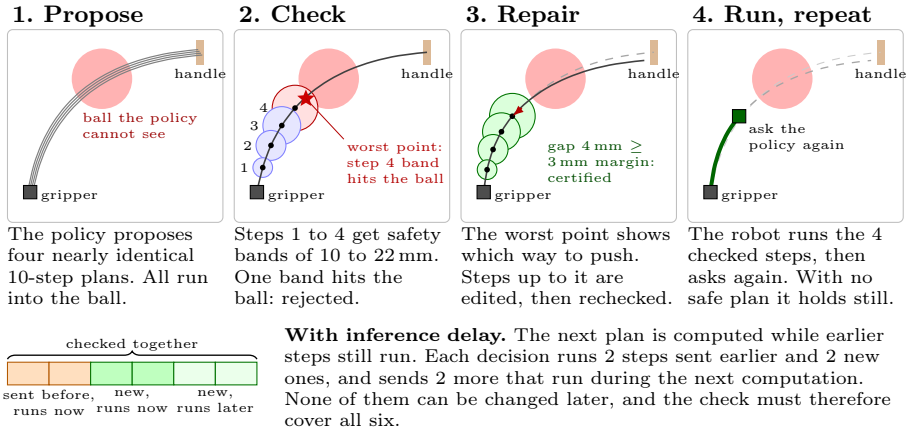


Fig. 1. One ShortStop decision, viewed from above with gripper bands only. Band widths and margin match the experiments. The scene and gaps are illustrative. Sections 3 and 4 formalize the steps.

sampling and contact dynamics, beyond the modest networks and short horizons supported by neural-network reachability tools [16,32].

Once proposed, however, a chunk has fixed actions. Their short-term effects under model error form a *reachtube* of possible arm positions. ShortStop sits between policy and robot as a *runtime shield*, checking this tube against constraints in Signal Temporal Logic (STL). The *robustness* of a constraint measures its margin of satisfaction. A failed check pinpoints the band nearest the obstacle. This *counterexample* guides repair, followed by another check. The loop draws on counterexample-guided synthesis and falsification [6,12,31]. The paper makes three contributions.

- A runtime shield checks and repairs chunks from a frozen generative policy (Sect. 3).
- The analysis establishes conditional safety, bounds unnecessary rejection, and derives the certification horizon required under inference delay (Sect. 4).
- Evaluation on 1000 pre-registered CALVIN obstacle scenes covers five baselines, two sharing our model, error bound and margins, with paired comparisons, an assumption audit, a delay test and ablations (Sects. 5–7).

2 Related Work

Generative policies. Diffusion produces action chunks [11, 29] or complete trajectories [19]. Flow matching generates actions through an ordinary differential equation [22]. Real-time chunking overlaps computation and execution [9]. Safety can enter generation through constraint projection [28], barrier conditions in the noise-removal network [34], or differentiable STL robustness [24]. In

ShortStop, the safety check follows generation and leaves the learned policy unchanged.

Safety filters. Shields can override unsafe actions using a verified state machine [2], and safe reinforcement learning constrains training [10]. Predictive safety filters optimize corrections to proposed inputs [33], and model-predictive shielding switches to a backup policy [8]. A control barrier function (CBF) inside a quadratic program (QP) keeps the state in a safe set [4], a property called *set invariance*. Hamilton–Jacobi reachability computes states from which safety can be maintained [7].

Closest to our setting, Reachability-Aided Imitation Learning (RAIL) filters diffusion actions [20]. PACS, a path-consistent safety filter, brakes along each generated path and checks its braking trajectory with reachable sets [30]. CBFs in learned representations filter image-based policies [25], and multi-link filters protect vision-language-action policies, which map images and language to actions [1]. Our focus within this setting is tube-based STL certification, counterexample-guided repair, and the horizon needed when computation overlaps execution.

Reachability, STL and counterexamples. Set-based motion bounds [3] and quantitative STL semantics [14,17] provide the basis for our online check [13]. Its witness connects monitoring to chunk revision through counterexamples [6, 12, 31].

3 Method

Problem setup. At discrete time t, the robot state x_t belongs to a state space $\mathcal{X}$ and evolves as $x_{t+1} = f(x_t, a_t) + w_t$. Here a_t is an action, f the unknown dynamics, and w_t a disturbance with norm bounded by $\bar{w}$. We write $\|\cdot\|$ for the Euclidean norm. Our frozen policy π_θ, with fixed learned parameters θ, maps observation o_t to a distribution over action chunks $a_{t:t+L} = (a_t, \ldots, a_{t+L-1})$. The slice excludes its endpoint, giving L *rows*, one action per step. The shield certifies H rows and executes r before the next decision, with $r \le H \le L$. For example, Fig. 1 proposes ten rows, checks four, and executes those four.

The experimental STL specification φ, formalized in (C), requires avoiding the unsafe set $\mathcal{X}_u$ at steps 1 to H. We write $\mathrm{dist}(x, \mathcal{X}_u)$ for the signed clearance between the arm at state x and the obstacles, negative under penetration.

(P) Proposal. We draw K candidate chunks $\{a^{(i)}\}_{i=1}^{K} \sim \pi_\theta(\cdot \mid o_t)$, where i indexes candidates, in one batched call (Fig. 1, panel 1). Their task-progress score $g(a)$ is the negative distance from the predicted end-effector (tool) position after the last row to the current sub-goal. Chunks that end closer to the goal rank higher. The simulator supplies the contact point and movement target, such as a drawer's handle and target opening position. Selection and repair act on chunks while leaving π_θ unchanged.

(R) Short-horizon reachtube. A band around each predicted arm pose accounts for prediction error. These sets $\hat{\mathcal{R}}_k$ at future steps k form the tube

$\hat{\mathcal{R}} = (\hat{\mathcal{R}}_1, \dots, \hat{\mathcal{R}}_H)$. Starting from the known state $\hat{\mathcal{R}}_0 = \{x_t\}$, we set

$$\hat{\mathcal{R}}_k = \hat{f}_k(x_t, a_{t:t+k}) \oplus \mathcal{B}(\epsilon_k + \bar{w}), \quad 1 \le k \le H, \tag{1}$$

where $\hat{f}$ is our approximate dynamics model, $\hat{f}_k$ its k-step prediction, and ϵ_k a calibrated bound on the k-step error. The Minkowski sum $\oplus\mathcal{B}(\varrho)$ adds every displacement in a ball of radius ϱ, expanding the prediction into a band. Because Eq. (1) bounds the full k-step deviation directly, ϵ_k must also cover accumulated disturbance. The implementation uses seven joint angles $q \in \mathbb{R}^7$ as its state. The tool Jacobian maps small joint motions to tool motions. Its damped least-squares inverse maps requested tool motions back to joint steps while limiting large updates [26]. Forward kinematics gives the corresponding arm positions, which we represent by *capsules*, line segments thickened by a radius. The hand capsule extends from the wrist flange to the tool point and covers the fingers at any opening. For collision checks, we represent $\hat{\mathcal{R}}_k$ in physical space by these capsules inflated by $\epsilon_k + \bar{w}$ (Fig. 1, panel 2).

(C) STL and its robustness over the tube. STL formulas in negation normal form follow the grammar

$$\varphi ::= \mu \mid \neg\mu \mid \varphi_1 \wedge \varphi_2 \mid \varphi_1 \vee \varphi_2 \mid \Box_I \varphi \mid \Diamond_I \varphi \mid \varphi_1 \,\mathcal{U}_I\, \varphi_2,$$

where a predicate μ states $h_\mu(x) \ge 0$ for a real function h_μ of the state, $I = [a, b]$ is an interval of steps with integers $0 \le a \le b$, and $\Box$, $\Diamond$ and $\mathcal{U}$ read "always", "eventually" and "until". For a trajectory $\xi = (\xi_0, \xi_1, \dots)$ with $\xi_0 = x_t$, the robustness at step k is defined recursively [14,17]:

$$\begin{aligned}
\rho(\mu, \xi, k) &= h_\mu(\xi_k), & \rho(\neg\mu, \xi, k) &= -h_\mu(\xi_k),\\
\rho(\varphi_1 \wedge \varphi_2, \xi, k) &= \min_j \rho(\varphi_j, \xi, k), & \rho(\varphi_1 \vee \varphi_2, \xi, k) &= \max_j \rho(\varphi_j, \xi, k),\\
\rho(\Box_I \varphi, \xi, k) &= \min_{k' \in k+I} \rho(\varphi, \xi, k'), & \rho(\Diamond_I \varphi, \xi, k) &= \max_{k' \in k+I} \rho(\varphi, \xi, k'),
\end{aligned}$$

$$\rho(\varphi_1 \,\mathcal{U}_I\, \varphi_2, \xi, k) = \max_{k' \in k+I} \min\Big(\rho(\varphi_2, \xi, k'),\ \min_{k \le k'' < k'} \rho(\varphi_1, \xi, k'')\Big),$$

and $\rho(\varphi, \xi) = \rho(\varphi, \xi, 0)$. Positive robustness implies that ξ satisfies φ, negative robustness implies violation, and $|\rho|$ measures how far the predicate values are from flipping the verdict. The interval semantics over a tube replaces each predicate value by its worst case over the band, $\rho(\mu, \hat{\mathcal{R}}, k) = \inf_{x \in \hat{\mathcal{R}}_k} h_\mu(x)$ and $\rho(\neg\mu, \hat{\mathcal{R}}, k) = \inf_{x \in \hat{\mathcal{R}}_k} (-h_\mu(x))$, and keeps the other rules. Because min and max are monotone, induction on φ gives $\rho(\varphi, \hat{\mathcal{R}}) \le \rho(\varphi, \xi)$ for every ξ with $\xi_k \in \hat{\mathcal{R}}_k$, which is Assumption 2. Avoidance takes the predicate $h(x) = \operatorname{dist}(x, \mathcal{X}_u)$ under $\Box_{[1,H]}$, and its tube robustness is

$$\rho(\varphi, \hat{\mathcal{R}}) = \min_{1 \le k \le H} \inf_{x \in \hat{\mathcal{R}}_k} \operatorname{dist}(x, \mathcal{X}_u), \tag{2}$$

The infimum finds the least clearance within each band, and the minimum selects the worst step. Certification requires $\rho(\varphi, \hat{\mathcal{R}}) \ge \gamma > 0$ for a chosen safety margin.

The 4 mm gap in Fig. 1, panel 3, passes our 3 mm margin. A 2 mm gap would fail despite leaving the nominal motion collision-free. Capsule–sphere distance has a closed-form expression, making this bound exact for the tube. Exact distances to other convex obstacles preserve this property. We evaluate only avoidance, leaving richer temporal specifications untested.

(S) Counterexample-guided repair and selection. Rejection supplies more than a verdict. It identifies the worst-case witness

$$(k^\star, x^\star) = \arg \min_{1 \le k \le H,\, x \in \hat{\mathcal{R}}_k} \operatorname{dist}(x, \mathcal{X}_u), \tag{3}$$

the closest point on the inflated arm (the star in Fig. 1, panel 2). If $\rho = \rho(\varphi, \hat{\mathcal{R}}) < \gamma$, the witness is a counterexample to certification. The gradient $v = \nabla_{a_{t:t+k^\star}} \operatorname{dist}(x^\star(a), \mathcal{X}_u)$, computed through the witness point's Jacobian, is the action direction that increases its clearance. With an extra repair margin $\mu > 0$, the Polyak step [27] makes the smallest change satisfying the linear approximation of the clearance constraint,

$$a_{t:t+k^\star} \leftarrow a_{t:t+k^\star} + \frac{\gamma + \mu - \rho}{\|v\|^2}\, v, \tag{4}$$

for nonzero v. The update changes only rows through $k^\star$ (panel 3), shifting the positions reached by later relative actions as well. Moving a band can expose a different closest point, requiring a new witness and a full tube check after each update, for at most n_{rep} rounds. If every original candidate fails, repair tries them in decreasing order of $g(a)$ until one certifies. Among the available certified chunks, we select

$$a^\star = \arg \max_{a \text{ certified}} g(a) \tag{5}$$

and execute its first r rows. If none certifies, our fallback controller π_{fb} holds the joints for one step before we query the policy again. Each decision needs K tube checks and at most Kn_{rep} repair rounds.

4 Theory

Assumption 1 (Sound tube). *For every committed chunk, the states that its first H rows would produce from x_t satisfy $x_{t+k} \in \hat{\mathcal{R}}_k$ for $1 \le k \le H$.*

Assumption 2 (Sound robustness bound). *For every tube and every trajectory ξ with $\xi_k \in \hat{\mathcal{R}}_k$ for $1 \le k \le H$, we have $\rho(\varphi, \hat{\mathcal{R}}) \le \rho(\varphi, \xi)$.*

Assumption 3 (Safe fallback). *There is a maximum clearance loss $0 \le \eta < \gamma$ such that, from any state with $\operatorname{dist}(x, \mathcal{X}_u) \ge \gamma$, holding with π_{fb} keeps $\operatorname{dist} \ge \gamma - \eta$ for any number of steps.*

The induction in Sect. 3 (C) establishes Assumption 2. Tube coverage and fallback safety require empirical evidence, examined in Sect. 6.2.

Theorem 1 (Short-horizon safety). *Let Assumptions 1 and 2 hold and let the shield commit at time t a chunk whose tube $\hat{\mathcal{R}}^\star$ satisfies $\rho(\varphi, \hat{\mathcal{R}}^\star) \geq \gamma > 0$. Then each state x_{t+k} with $1 \leq k \leq H$ that the chunk's first H rows would produce satisfies $\mathrm{dist}(x_{t+k}, \mathcal{X}_u) \geq \gamma$. In particular the $r \leq H$ executed rows keep the arm at least γ from $\mathcal{X}_u$.*

Proof. Assumption 1 places the states inside $\hat{\mathcal{R}}^\star$. Assumption 2 bounds their robustness below by γ. For avoidance, Eq. (2) therefore gives clearance at least γ at every checked step. The executed prefix has $r \leq H$ rows, and repaired chunks pass the same check. □

Corollary 1 (All-time safety). *Suppose Assumptions 1–3 hold, with a sound tube at every decision. If the initial state x_0 satisfies $\mathrm{dist}(x_0, \mathcal{X}_u) \geq \gamma$, then every state has clearance $\mathrm{dist}(x_t, \mathcal{X}_u) \geq \gamma - \eta > 0$.*

Proof. Every executed certified row ends with clearance at least γ by Theorem 1, which does not require the chunk to start from a safe state. A stall therefore starts with clearance at least γ, and Assumption 3 keeps dist $\geq \gamma - \eta$ until the next certified chunk runs. □

Tube coverage is statistical in practice because ϵ_k is estimated from data. Let e_k be the largest position error after k rows over the nine arm frame origins and the tool point, measured as $e_k^{(1)}, \ldots, e_k^{(n_k)}$ on n_k calibration decisions. *Exchangeability* means that the joint distribution of calibration and deployment errors is unchanged by reordering.

Corollary 2 (Statistical tube soundness). *Let j_k be a rank fixed before calibration and let ϵ_k be at least the j_k-th smallest calibration value. If the deployment value e_k is exchangeable with the calibration values, then $\Pr[e_k > \epsilon_k] \leq \delta_k := 1 - j_k/(n_k + 1)$, where Pr denotes probability. This escape bound δ_k is marginal over calibration and deployment. Assumption 1 holds at a decision with probability at least $1 - \sum_{k=1}^{H} \delta_k$.*

Proof. Exchangeability gives the rank-based coverage of at least $j_k/(n_k+1)$ used in split-conformal prediction [5]. Every capsule spans two measured points and interpolates linearly between them. Hence $e_k \leq \epsilon_k$ for all $k \leq H$ places the arm inside $\hat{\mathcal{R}}_k$. A union bound over k completes the proof. □

No independence assumption is needed for the union bound. Its *marginal* coverage need not hold for every selected chunk. Across N decisions, the all-time bound weakens to $1 - N \sum_k \delta_k$.

Remark 1 (Certified horizon under inference delay). With an asynchronous executor [9], computation overlaps motion. Let $d < r$ be the inference delay in action steps. Each decision executes the d rows that the previous decision committed and the next $r - d$ rows of the new chunk. It also commits the following d rows, which run while the next chunk is computed and which the next decision can no longer change. A certificate must cover all of these rows, which gives

$$H \geq r + d \tag{6}$$

counted from x_t, with the committed rows held fixed during repair. With $r = 4$ and $d = 2$ this is six rows (Fig. 1, bottom), and $H = r$ leaves d committed rows uncertified.

Conservatism and horizon. Error bands can reject nominally safe motions, a cost called *conservatism*. A longer horizon widens the bands that candidates must clear. Let $\varrho_H = \epsilon_H + \bar{w}$ be the tube radius at the last certified step, let ρ_{nom} be the robustness (2) of the nominal prediction ($\epsilon_k = \bar{w} = 0$), and let $\alpha(H) = \Pr[\rho(\varphi, \hat{\mathcal{R}}) < \gamma]$ be the rejection rate over the policy's chunk and state distribution.

Proposition 1 (Conservatism bound). *Suppose the probability density of ρ_{nom} is at most β on $[0, \gamma + \varrho_H]$. Then*

$$\alpha(H) \le \alpha^\star + \beta\,(\gamma + \varrho_H), \tag{7}$$

where $\alpha^\star = \Pr[\rho_{\text{nom}} < 0]$ is the rate of chunks whose nominal prediction violates φ.

Proof. Each band lies within ϱ_H of the nominal arm, hence $\rho \ge \rho_{\text{nom}} - \varrho_H$ and $\{\rho < \gamma\} \subseteq \{\rho_{\text{nom}} < \gamma + \varrho_H\}$. Splitting this event at zero gives (7). □

The calibrated radii grow approximately as $6.7 + 3.8k$ mm (9.6 to 36.7 mm for $k = 1, \ldots, 8$), motivating the study of look-ahead costs in Sect. 7.

5 Experimental Setup

Benchmark and policy. We use CALVIN (Composing Actions from Language and Vision) [23], a simulated tabletop benchmark with a seven-degree-of-freedom Franka arm controlled at 30 Hz. All methods use environment D and the public checkpoint of MDT (Multimodal Diffusion Transformer) trained on environments A–D [29]. MDT emits chunks of $L = 10$ relative end-effector actions after 10 denoising steps. Each *rollout* attempts a language-specified task from a fresh initial state, with success determined by CALVIN's task oracle and a limit of 360 steps. The fresh-start requirement leaves 27 tasks, excluding stacking, placing and lifting from the drawer (7 of 34 tasks). The policy remains frozen throughout.

Obstacles. Each scene adds 1–3 spheres (radius 3–7 cm, at least 2 cm apart) within 20 cm of an anchor near the task's contact goal or on the arm's reference path. The policy never sees them. A scene is kept only if (i) the start pose clears every sphere by at least 5 cm, (ii) the spheres stay clear of the task's contact and completion targets, (iii) 20–80% of reference unshielded rollouts touch them, and (iv) an RRT-Connect planner [21] finds a start, contact and completion path with a 2 cm margin. Thus every retained scene has a collision-free kinematic path. A rollout ends on success, at the first collision between an arm or finger capsule and a sphere, or at the step limit.

Shield knowledge. The shield knows the exact sphere centres and radii, and $g(a)$ reads the task's contact and completion targets from the simulator.

The policy sees neither. Both the model predictive control (MPC) filter and CBF receive the same obstacle geometry, but neither uses $g(a)$. Section 7 measures the contribution of $g(a)$ without repair. Its effect on repair order remains unisolated. Deployment would require obstacle perception and a progress proxy that works without a task oracle.

Splits and protocol. Tuning is restricted to 40 validation scenes. For testing, we freeze 1000 scenes over 97 previously unused initial states, recording each task, state, obstacle layout and policy seed. Each method runs once per scene using the same seed, yielding paired outcomes. A timestamped lab notebook, released with the code, registers the configuration, horizon-selection rule, and delay and ablation studies beforehand. Tasks are interleaved with randomly drawn states. The first 200 and 300 scenes cover all 27 tasks and 89 and 96 initial states.

ShortStop instantiation. We set the Jacobian inverse's damping parameter to $\lambda = 0.03$. For ϵ_k, we take the 99th percentile of e_k on obstacle-free rollouts of 50 task sequences (2489 to 2695 decisions per k), multiplied by 1.25 (9.6, 14.7, 18.5, 22.4, ..., 36.7 mm for $k = 1, \ldots, 8$). With j_k the rank of that percentile, δ_k is 1.04 to 1.08% for $k = 1, \ldots, 4$, and Corollary 2 gives a per-decision soundness probability of at least 95.7% for $H = 4$. Only 0.12 to 0.26% of calibration errors exceed ϵ_k. In the deterministic simulator, we set $\bar{w} = 0$ and absorb all error, including controller tracking lag, into ϵ_k. The margin is $\gamma = 3$ mm, and further parameters are $K = 4$, $r = 4$, $n_{\text{rep}} = 5$ and $\mu = 1$ mm. We selected $H = 4$ on validation from $\{4, 6, 8\}$ by the registered rule, which picks the highest mean safe success over the three model-based filters. During a stall, the fallback holds the joints at their starting angles. The measured drift is at most 0.25 mm over 300 held steps, supporting Assumption 3 with $\eta = 0.25$ mm for stalls of that length. Our longest observed stall lasts 340 steps. Apart from H and the choice $K = 4$ after a probe of $K \in \{1, 4, 8\}$ on validation, nothing in ShortStop was tuned.

Baselines. MPC-Filter and CBF use the identical model, capsule geometry, constraint tightening (adding ϵ_k to the required clearance), 3 mm margin and hold fallback. STL-Monitor shares the model, geometry and fallback. *Unshielded* executes the policy's chunk. *Conf-Thresh* rejects candidates whose predicted end-effector endpoints lie more than 5 mm (tuned on validation) from their centroid (mean endpoint) and picks the best $g(a)$ among the rest, without any obstacle knowledge. *STL-Monitor* [13] certifies the nominal STL robustness of the model rollout ($\epsilon_k = 0$, no tube) with a 1 cm margin, picks the best $g(a)$ among accepted candidates and has no repair. *MPC-Filter* [33] minimizes the deviation from the policy's chunk over H steps under the tightened collision and joint-limit constraints using sequential quadratic programming (SQP), which solves at most 5 local QPs. *CBF* [4] filters each executed row with a discrete-time barrier QP, $h(q_{k+1}) \geq (1 - \kappa)\, h(q_k)$, with joint limits. Here $h(q)$ is the tightened capsule–sphere clearance, and $\kappa = 0.5$ (tuned on validation) limits its shrinkage per step. Infeasibility triggers the fallback for every method.

Metrics. Per rollout we record *safe success* (task completed with no collision), *violation* (any collision) and *timeout* (step limit reached without success

or collision). Their rates sum to one. Per decision we record *activation* (the proposed chunk was not run as proposed) and *fallback*. Collisions are checked at the 30 Hz samples, like the certificate, and motion between samples (up to 2 cm) is not covered. *Conservatism cost* is the success lost relative to the unshielded policy on the 603 scenes in which the unshielded policy does not collide. *Added latency* per decision is the shield's computation (selection, repair, solvers, and $g(a)$ for the methods that rank by it), measured after a warm-up decision. All intervals are 95% bootstrap confidence intervals (CIs), obtained from 10^4 resamples of scenes [15]. Two-sided paired bootstrap tests keep each scene's outcomes together across methods. A scene is *discordant* when two methods produce different outcomes. The ten comparisons of ShortStop with each baseline on safe success and violation form the primary family, which we correct with Holm's procedure [18]. Secondary analyses report uncorrected p-values. With 10^4 resamples the smallest resolvable p is $2 \cdot 10^{-4}$, and we report $p < 0.001$ below that. All runs use one consumer laptop (Intel i5-12450HX, RTX 3050 6 GB for the policy), with shields using one central processing unit (CPU) thread.

Table 1. Main results on 1000 CALVIN test scenes (one rollout per scene, same policy seed for every method). Brackets: 95% bootstrap CIs. Bold: best value, significant or not. Cons. cost: conservatism cost in percentage points. Latency: added shield computation per decision, with median (med), 95th/99th percentiles (p95/p99), and maximum.

	Safe succ.	Violation	Time-	Activ.	Fallb.	Cons.	Latency (ms)	
Method	(%) ↑	(%) ↓	out (%)	(%)	(%)	cost ↓	med / p95	p99 / max
Unshielded	58.4 [55.3, 61.5]	39.7 [36.6, 42.8]	1.9	–	–	–	–	–
Conf-Thresh	59.2 [56.2, 62.3]	39.2 [36.1, 42.2]	1.6	1.6	0.0	2.7	37 / 61	71 / 121
STL-Monitor	39.8 [36.8, 42.8]	**0.0** [0.0, 0.0]	60.2	93.6	93.6	32.2	48 / 71	80 / 170
MPC-Filter	78.5 [76.0, 81.0]	1.0 [0.4, 1.7]	20.5	49.6	5.6	2.7	13 / 291	652 / 1357
CBF	**80.3** [77.8, 82.8]	0.7 [0.2, 1.3]	19.0	50.4	2.7	2.5	101 / 206	230 / 309
ShortStop	80.0 [77.5, 82.5]	1.1 [0.5, 1.8]	18.9	52.9	11.1	**1.8**	56 / 148	328 / 408

6 Results

6.1 Main Comparison

Shielding improves both collision avoidance and task completion (Table 1). Collisions fall from 39.7% under unshielded execution to 1.1% (−38.6 percentage points, $p < 0.001$), while safe success rises from 58.4% to 80.0% (+21.6 points, $p < 0.001$). These gains describe scenes deliberately chosen to challenge the unshielded policy through filter (iii). All filters face the same scenes.

In paired outcomes, ShortStop alone succeeds in 235 scenes and the unshielded policy alone in 19. Of the 397 scenes with unshielded collisions, ShortStop completes 57.2% safely (CBF 58.9%, MPC 54.7%). Most of the remaining scenes time out. Intervention is frequent, affecting 52.9% of proposals, but only 11.1% of decisions end in a hold.

Task and clearance differences. Contact requirements help explain task variation (Fig. 2a). Pushing requires sustained contact with a moving block near

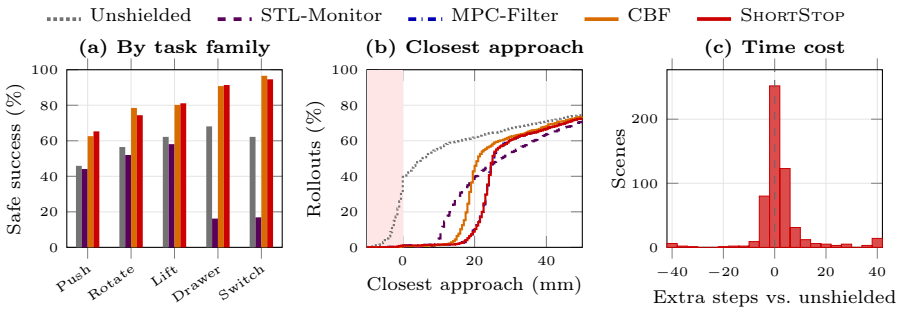


Fig. 2. Three views of the main run that Table 1 does not show. (a) Safe success per task family (MPC-Filter, omitted, is within 4 points of CBF). (b) Cumulative share of rollouts by closest approach of the real arm to an obstacle (shaded: collision). (c) ShortStop minus unshielded steps on the 557 scenes all four methods complete, clipped at ±40 steps.

obstacles, limiting safe success to 62–65% across the three model-based filters. Drawer and switch tasks reach 90–97%. The drawer family includes drawer, slider and push-into-drawer tasks, while switches operate a light-emitting diode (LED) or lightbulb. STL-Monitor completes just 16% and 17% of these two families: repeated rejection can prevent the contact needed to finish. Neither ShortStop nor CBF leads on every task. Pushing favours ShortStop (65.3% against 62.6%), whereas rotation favours CBF (78.5% against 74.4%).

The filters also keep the arm farther from obstacles (Fig. 2b). With bands of 10–22 mm around the predicted arm and a further 3 mm margin, the median closest approach is 25 mm for ShortStop and MPC, or 21 mm for CBF. Fewer than 2% of their rollouts pass within 10 mm of an obstacle, compared with 56% unshielded. This extra clearance costs little execution time on the 557 scenes all four methods complete safely: each has a median completion time of 66 steps. For ShortStop, the paired difference from unshielded is zero at the median and within 10 steps in 90% of scenes (Fig. 2c). The main cost appears in scenes that stall.

Comparison with model-based filters. With the model, error bound, margin and fallback held fixed, the three filters perform similarly. The violation rate for ShortStop differs from CBF and MPC by +0.4 and +0.1 points ($p = 0.21$ and 0.88). For safe success, the difference from CBF is −0.3 points ($p = 0.76$, 33 vs. 36 discordant scenes), while the +1.5 points over MPC ($p = 0.0496$) do not survive Holm's correction. All $p < 0.001$ comparisons here remain significant after correction. Conservatism cost follows the same pattern: our estimate is 1.8 points [0.2, 3.5], CBF's is 2.5 [1.2, 3.9], and MPC's is 2.7 [1.3, 4.1], with neither difference statistically significant ($p = 0.43$ and 0.34). Thus the measured outcomes are comparable despite different correction rules.

STL-Monitor exposes the cost of frequent rejection. It records no collisions, but rejects 93.6% of decisions and leaves 60.2% of scenes unfinished. Relative to this monitor, ShortStop gains 40.2 safe-success points ($p < 0.001$, 407 vs. 5 discordant scenes) with 1.1 points more violations. Differences in tube, margin and

Table 2. Component ablation on the first 300 test scenes ($H = 4$). Each rung adds one component. Last column (Δ): paired safe-success change against the rung above (points, p). Lat.: p95 latency in ms over these 300 scenes (Table 1 pools all 1000).

Rung	Configuration	Safe succ. (%)	Viol.	Activ.	Fallb.	Lat.	Δ vs. above
L0	MDT policy, no shield	56.3 [51.0, 62.0]	41.7	–	–	–	–
L1	+ certify (tube, margin) + fallback, K=1	27.7 [23.0, 33.0]	0.0	95.2	95.2	28	−28.7 (<.001)
L2	+ K=4 candidates (policy order)	27.7 [22.7, 33.0]	0.0	95.2	95.2	46	+0.0 (1.000)
L3	+ progress score $g(a)$	28.7 [23.7, 34.0]	0.0	95.2	95.2	77	+1.0 (.401)
L4	+ counterexample repair (ShortStop)	79.3 [74.7, 83.7]	0.7	55.1	14.6	253	+50.7 (<.001)

repair prevent attribution to repair alone, which Sect. 7 isolates. Conf-Thresh, meanwhile, behaves much like the unshielded policy: disagreement among samples fails to flag unsafe motions here.

Latency and continued progress. Similar outcomes conceal different computational costs. ShortStop adds a median 56 ms per decision, including 27 ms for the progress score, with p99 328 ms and maximum 408 ms. MPC handles feasible chunks quickly (median 13 ms), but its p99 reaches 652 ms and its maximum 1.36 s. CBF varies less (median 101 ms, maximum 309 ms), with one QP solved per executed row in our Python implementation. MDT inference adds a median 104 ms on our graphics processing unit (GPU). The measured median full decision takes about 165 ms, exceeding the 133 ms available for $r = 4$ steps at 30 Hz. This mismatch makes inference delay part of the execution problem (Remark 1). A feasible QP can often produce a small motion when our repair fails, helping explain the higher fallback rate of ShortStop (11.1% vs. MPC's 5.6% and CBF's 2.7%). Despite this difference, all three time out in 19–21% of scenes.

6.2 Empirical Soundness of the Certificate

A passive audit compares predicted bands with actual motion on the first 300 test scenes. It reproduces all 300 main-run outcomes while examining tube coverage and fallback safety (Assumptions 1 and 3). Across 41,899 certified rows, the arm leaves its tube in 0.10% (0.06, 0.08, 0.11 and 0.17% at $k = 1, \ldots, 4$), below the δ_k levels in Sect. 5. The 99th-percentile errors, 6.0–14.4 mm, remain below their corresponding radii. Actual robustness falls below the certificate in 0.02% of rows. Assumption 2 holds by construction. Across all 1000 scenes, none of our 11, MPC's 10 or CBF's 7 collisions occurs during fallback.

The rare failures reveal a specific weakness. Both collisions in the audited subset occur during certified motion, when pushing or lifting produces errors of 22 and 51 mm against radii of 14.7 and 18.5 mm. Contact also dominates failures in the full test set: lifting a block from the slider accounts for 9 of our 11 collisions and 6 of CBF's 7. These contact effects were absent from the obstacle-free calibration.

Table 3. Inference delay d on the first 200 test scenes (%). Naive: H=4. Aware: H=4+d (CBF: rows after the committed ones). Last two columns: paired difference in points with bootstrap p.

		Safe succ. / viol.		Aware − naive (p)	
d	Method	Naive	Aware	Safe succ.	Viol.
1	Unshielded	54.5 / 43.5	–	–	–
	ShortStop	77.0 / 2.0	74.0 / 0.5	−3.0 (.107)	−1.5 (.247)
	MPC-Filter	78.5 / 3.0	75.0 / 0.0	−3.5 (.052)	−3.0 (.003)
	CBF	80.5 / 1.5	78.5 / 1.5	−2.0 (.200)	+0.0 (1.000)
2	Unshielded	53.5 / 45.5	–	–	–
	ShortStop	80.0 / 14.0	74.5 / 0.5	−5.5 (.033)	−13.5 (<.001)
	MPC-Filter	78.5 / 15.5	72.5 / 0.5	−6.0 (.027)	−15.0 (<.001)
	CBF	77.5 / 13.5	75.5 / 1.5	−2.0 (.298)	−12.0 (<.001)
3	Unshielded	55.0 / 44.5	–	–	–
	ShortStop	68.5 / 31.5	72.0 / 0.0	+3.5 (.330)	−31.5 (<.001)
	MPC-Filter	67.5 / 32.5	68.5 / 0.0	+1.0 (.825)	−32.5 (<.001)
	CBF	70.0 / 29.0	74.0 / 0.5	+4.0 (.216)	−28.5 (<.001)

6.3 Inference Delay

We test Remark 1 on the first 200 test scenes with $d \in \{1, 2, 3\}$, where the next chunk is computed while the committed rows run [9]. Each filter runs in a *naive* mode (the final configuration with $H = 4$, unaware of the delay) and in an *aware* mode ($H = 4 + d$, committed rows fixed). CBF has no horizon, and its aware mode leaves the committed rows uncorrected and filters the rows after them. Nothing is re-tuned (Table 3).

Committed rows become consequential once the delay reaches $d \geq 2$: all three filters benefit substantially from accounting for them (Table 3). Without this adjustment, violations under ShortStop rise from 2.0% to 14.0% to 31.5% as d increases, approaching the unshielded rate of 44.5%. Certifying $H = r + d$ keeps the corresponding rates at 0.5%, 0.5% and 0.0%. The reductions at $d = 2$ and 3 are 13.5 and 31.5 points ($p < 0.001$). MPC and CBF show the same pattern, with aware violation rates at most 1.5%. Longer certification costs 2–6 safe-success points at $d \leq 2$. At $d = 3$, where collisions dominate the naive mode, the aware mode has higher safe success (72.0% vs. 68.5%, $p = 0.33$). Our aware safe success ranges from 72.0% to 74.5%, within ±4.5 points of the baselines.

7 Analysis and Ablations

To separate repair from the other design choices, we build the shield one component at a time on the first 300 scenes (Table 2). Starting with the unshielded policy (L0), we certify one sample and hold on rejection (L1), then draw $K = 4$ samples and take the first certified one (L2). Ranking by $g(a)$ gives L3. Adding repair yields the full method, L4.

Certification alone removes the observed collisions: L1 reduces violations from 41.7% to 0.0%. But the robot holds in 95% of decisions, times out in 217 of 300 scenes, and completes only 27.7% safely. Sampling more chunks changes little (L2, +0.0 points, 1 vs. 1 discordant scene), as does ranking them (L3, +1.0

point, 6 vs. 3). MDT's samples are nearly identical and tend to fail the same check. Repair changes the motions available for execution, adding 50.7 safe-success points (152 scenes won, none lost, $p < 0.001$) and reducing fallback from 95.2% to 14.6%. The two collisions introduced in L4 (0.7%) are the contact cases discussed in Sect. 6.2.

Horizon. Without delay, certifying beyond the executed prefix offers more look-ahead but requires wider bands. We compare $H \in \{4, 6, 8\}$ on the same 300 scenes, covering the executed rows while staying within our calibrated range. Validation safe success is 75.0, 72.5 and 72.5%. On test, extending H to 6 and 8 reduces safe success to 73.3% and 70.7%, losses of 6.0 and 8.7 points ($p < 0.001$ each, 18 vs. 0 and 29 vs. 3 discordant scenes). Violations remain within one scene (0.7, 0.3 and 0.7%). Fallback rises from 14.6% to 18.4% and 15.9%, and p95 latency from 253 to 286 and 299 ms. These results are consistent with the conservatism bound in Proposition 1. Among the tested settings, $H = r$ works best without delay, while the delay study supports $H = r + d$.

8 Discussion and Limitations

Benchmark scope. Static-obstacle avoidance reduces our STL check to set invariance, a setting well suited to a CBF. Comparable performance here leaves open whether richer temporal specifications would favour our formulation. The low proposal diversity also matters: MDT endpoints lie only 0.7 mm from their centroid at the median, limiting what additional samples can offer. Our kinematic model captures arm geometry but omits the contact effects associated with residual collisions.

All experiments use one simulator, one policy and one rollout per scene. The 1000 paired scenes leave policy-seed variation unmeasured. Hardware evaluation and comparison with PACS [30], which brakes along the proposed path, remain to be done. The deterministic simulator also leaves nonzero $\bar{w}$ untested.

Implementation limits. Joint limits are checked on the nominal path but omitted from our obstacle-repair objective. MPC and CBF include them in their QPs. Affected decisions (3%) fall back, costing at most 0.4 safe-success points. Repairs and contact can break exchangeability, while long episodes weaken the union bound. The observed escape rate of 0.10% concerns sampled states, leaving motion between 30 Hz samples uncovered. The 11% fallback rate also limits *liveness*, or continued task progress.

9 Conclusion

ShortStop uses failed clearance checks to repair the short-term motions of a frozen generative policy. Its safety guarantee depends on tube coverage, observed in 99.9% of audited steps. On CALVIN, collisions fall by 97% relative to unshielded execution and safe success rises by 21.6 points, with outcomes comparable to filters sharing our model. The delay study yields a practical requirement: certification must follow the execution schedule. Accounting for the

$H \geq r + d$ rows keeps every tested filter's violation rate at or below 1.5%. Temporal specifications, comparison with PACS, and hardware evaluation are the next tests of how far these findings extend.

Code and data availability. Code, frozen scenes, calibration data and all rollouts are available at https://github.com/congtul/shortstop-calvin.

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